

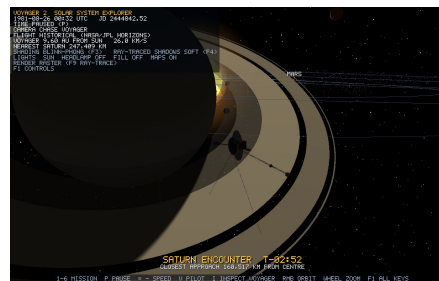
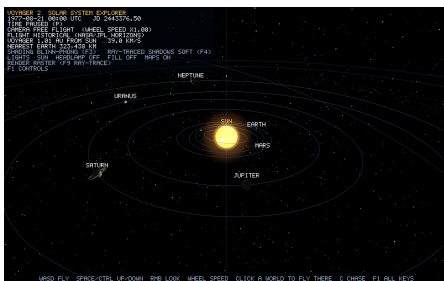
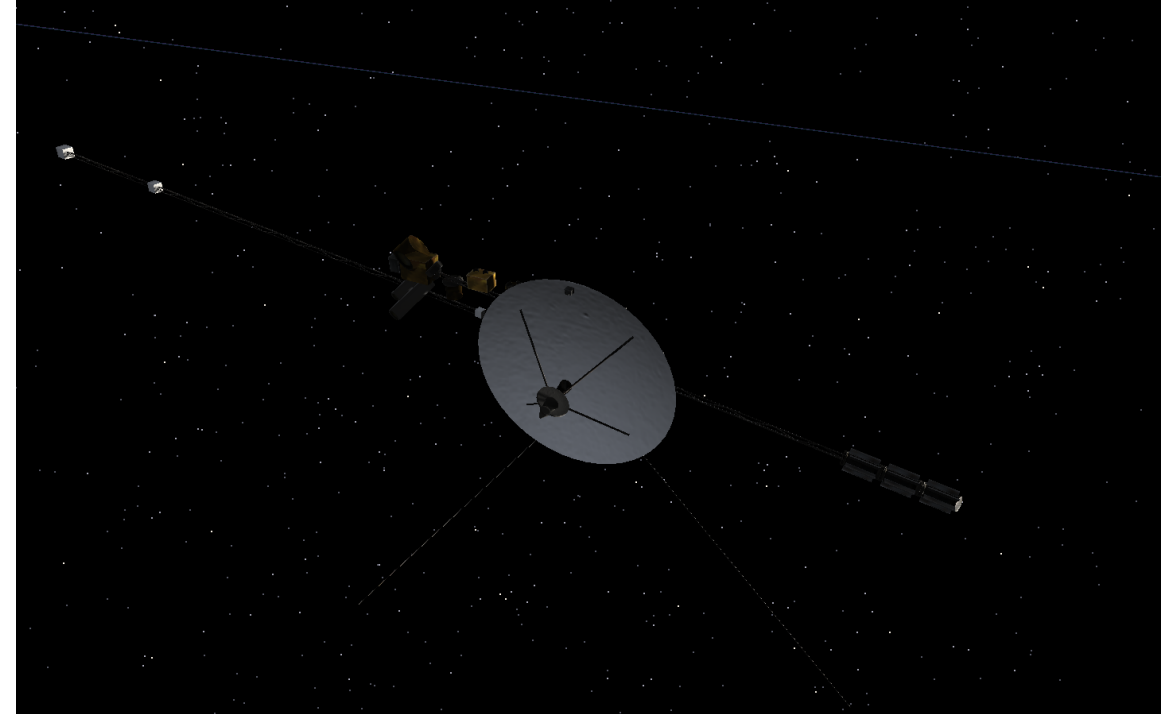
# Voyager 2 Solar-System Explorer

01 INTRODUCTION · What the project is and why it was built

**Explore a 3D solar system.**  
**Explain how it is rendered.**

Replay flybys and inspect the spacecraft.  
Fly manually. Compare rendering techniques.

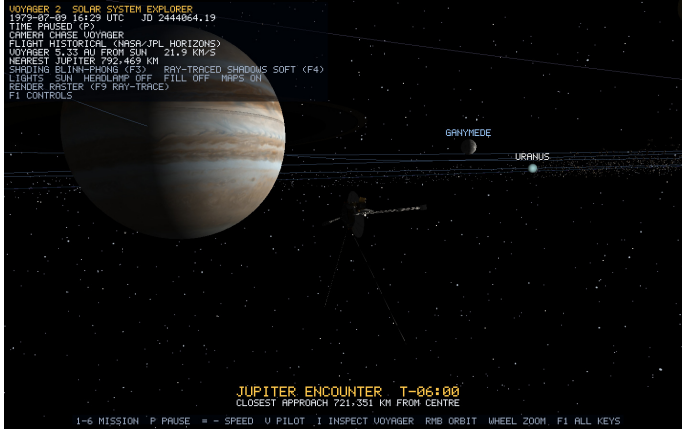
**26 textured bodies · 4 ring systems**  
**76 assemblies · 18 hardware targets**  
**8,500 instanced rocks · dynamic comet**



**Sarwad Hasan Siddiqui | Roll:**  
**2107006**  
**CSE 4102 | KUET | October 2026**

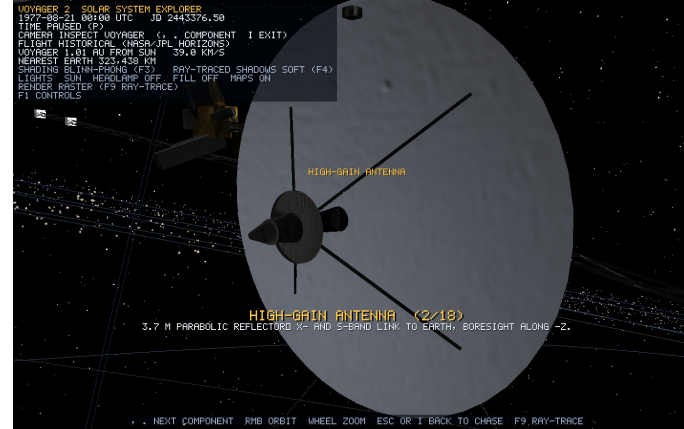
# Explore, control, compare

02 INTRODUCTION · A complete project walkthrough, followed by your recorded video



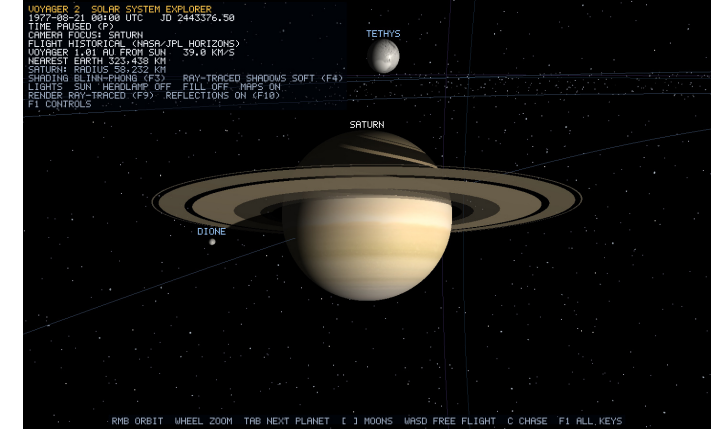
## MISSION

Dated launch, four flybys and heliopause.



## INSPECTION

Body focus, 18 hardware targets, free camera.



## COMPARISON

Three lights, five shading modes, ray tracing.

**C++20 · OpenGL 3.3 · GLFW / GLAD / GLM**

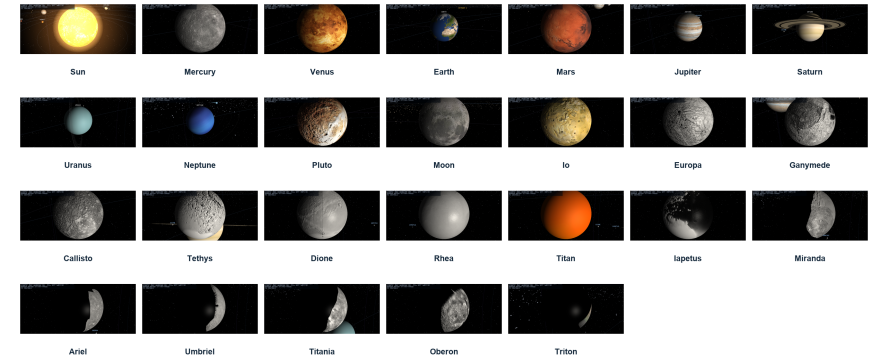
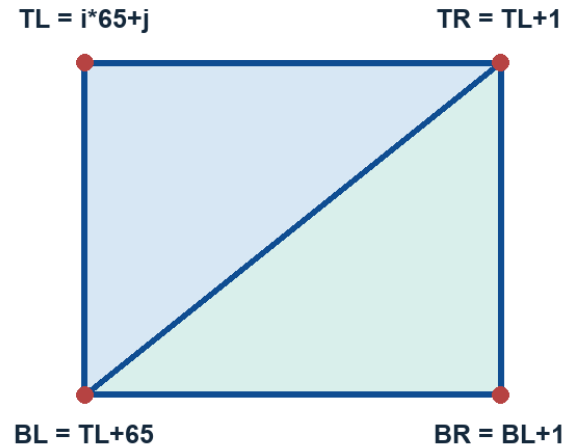
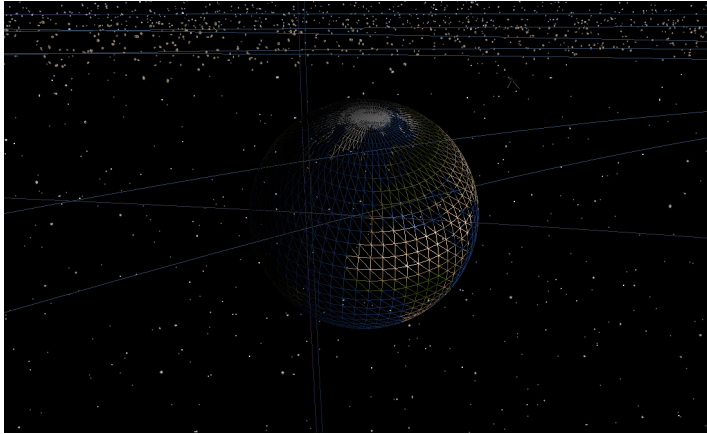
Course teachers: Md Tajmilur Rahman  
and Md Mubtashim Abrar Nihal — Lecturers

**PLAY PROJECT VIDEO**

Attach your final recording here.

# Vertices → indices → triangles → worlds

03 GEOMETRY · Shared sphere construction creates all 26 textured bodies



**Vertex = position + normal + UV**

$$\mathbf{p} = (\sin \phi \cos \theta, \cos \phi, \sin \phi \sin \theta)$$

$$\phi = \pi i / L, \quad \theta = 2\pi j / N, \quad uv = (1 - j / N, 1 - i / L)$$

Cell triangles: (TL, TR, BL) and (TR, BR, BL).

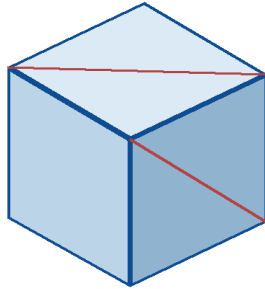
Poles omit degenerate halves; the UV seam duplicates positions.

**32 × 64 shared sphere**  
**2,145 vertices · 3,968 triangles**  
**11,904 indices · 26 materials/maps**

# Voyager: authored from reusable primitives

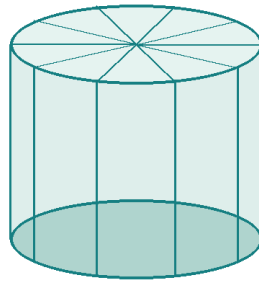
04 OBJECT CONSTRUCTION · Boxes, cylinders, frustums, rods and a curved parabolic dish

BOX / HARD EDGES



24 vertices / 12 triangles

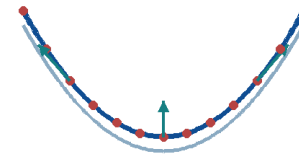
CYLINDER / FRUSTUM



$4n+6$  vertices /  $4n$  triangles

PARABOLIC DISH

48 angular segments / 6 radial rings



$$y = -D + D(r/R)^2$$

590 vertices / 1,152 triangles

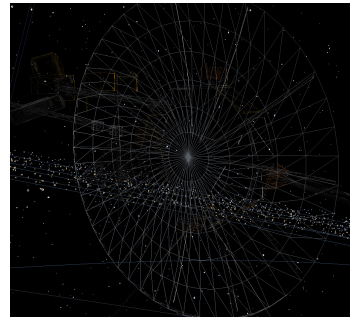
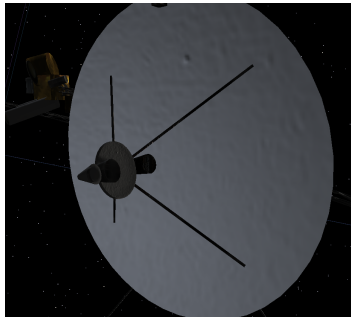
**BOX: 24 vertices / 12 triangles**

**CAPS + SIDES:  $4n+6$  /  $4n$**

**DISH: 590 / 1,152**

$$y = -D + D(r/R)^2$$

$$\mathbf{n} \propto (-2Dx/R^2, 1, -2Dz/R^2)$$



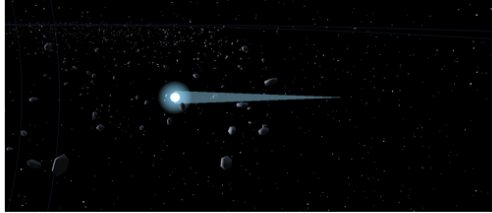
**76 assemblies · 5,828 triangles**

**18 inspectable hardware components**

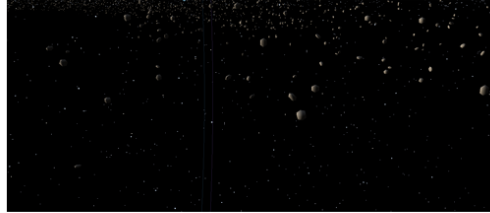
**Rods: rotate +Y onto B-A; rebase indices.**

# Rings, belts, stars, comet and scene guides

05 OBJECT FAMILIES · Every rendered family has a primitive and a construction rule



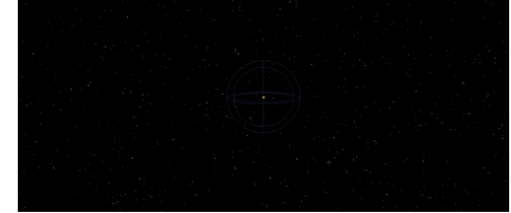
Dynamic comet



Asteroid belt / 4,000



Kuiper belt / 3,000



Oort shell / 1,500

## RINGS

$$\mathbf{p} = (r \cos\theta, 0, r \sin\theta)$$

Inner/outer edges; reversed back face.

260 vertices / 256 triangles per band.

## ROCK FIELDS

28 vertices / 24 triangles per rock.

8,500 matrices  $\rightarrow$  3 instanced draws.

Deterministic placement; static fields.

## STARS / GUIDES

7,380 points; 96-vertex line circles.

No surface triangles for these guides.

HUD text uses screen-space quads.

**DYNAMIC COMET = sphere nucleus + glow coma + 16-segment cone tail**

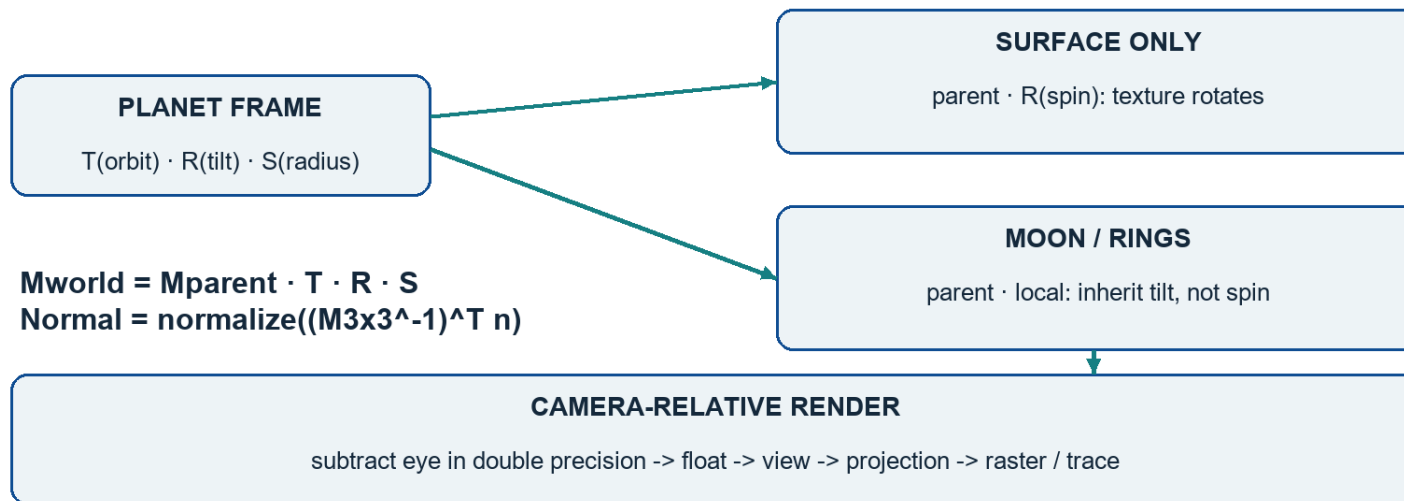
$$\mathbf{d}_{tail} = \text{normalize}(\mathbf{p}_{comet} - \mathbf{p}_{Sun})$$

52 cone vertices / 32 triangles

# Transforms, hierarchy, scale and cameras

06 TRANSFORMATIONS · Position, pose, size and view are separate operations

$$\mathbf{p}_{clip} = PVM_{world}\mathbf{p}, \quad M_{world} = M_{parent}TRS$$



$$\mathbf{n}' = \text{normalize}((M_{3 \times 3}^{-1})^T \mathbf{n})$$

Surface-only spin · parent compensation · quaternion rigs · logarithmic depth

## Four camera cases

FreeFly: independent motion

Focus: body-centred orbit

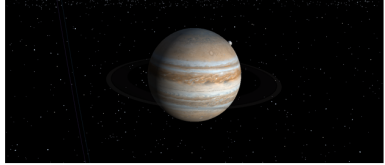
Chase: spacecraft frame

Inspect: hardware close-up

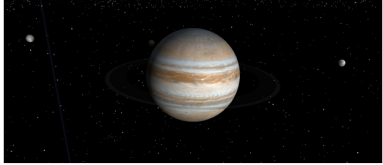
Subtract the eye in double precision.  
Compressed scale keeps bodies visible;  
physical AU/km values remain in the HUD.

# Every motion has an owner and a clock

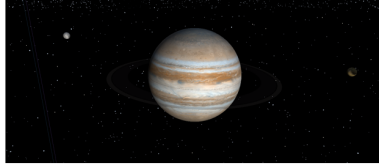
07 MOTION · Dated flight, moon revolution, axial spin, manual six-DOF and comet orientation



Jupiter / initial



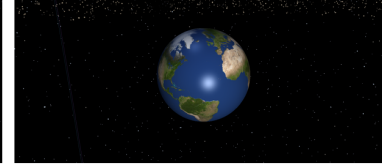
Moons / later



Moons / later still



Spin / initial



Spin / later



Spin / later still

## MOON REVOLUTION / tilted parent frame

$$\mathbf{p}(s) = h_{00}\mathbf{p}_0 + h_{10}h\mathbf{v}_0 + h_{01}\mathbf{p}_1 + h_{11}h\mathbf{v}_1$$

$$r = \frac{a(1 - e^2)}{1 + e \cos \nu}, \quad \dot{\nu} = 2\pi(0.4)/P_{days}$$

$$q' = \text{normalize}(qq_y q_x q_z)$$

## AXIAL SPIN / surface transform only

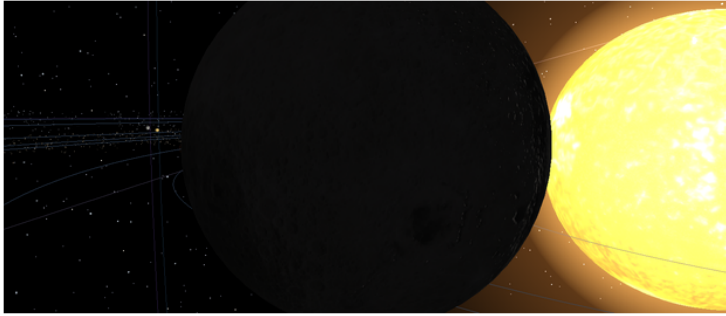
$$\dot{\psi} = 2\pi/P_{hours}$$

$$\mathbf{v}' = \mathbf{v} + \mathbf{a}\Delta t, \quad \mathbf{p}' = \mathbf{p} + \mathbf{v}\Delta t$$

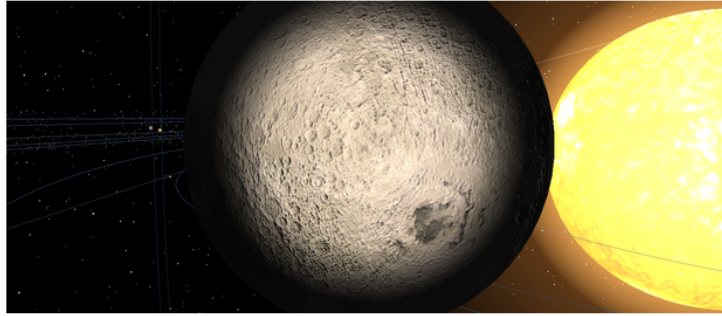
**Historical data: one date. Visual moons/spin: independent rates.**  
**Manual flight + cameras: real time. Comet tail: anti-Sun direction.**

# Three light types—and a moving headlamp

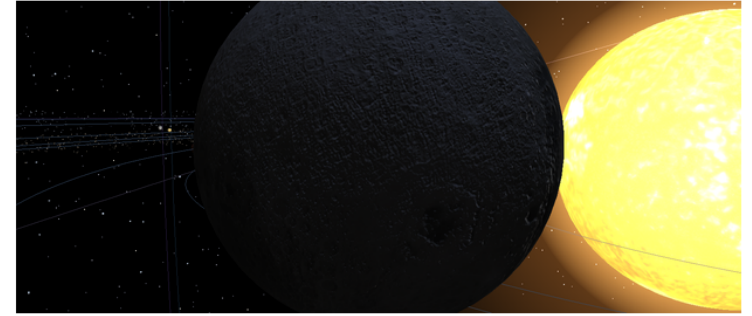
08 LIGHTING · Source geometry and attenuation determine incident illumination



Sun point only



Point + camera spotlight



Point + directional fill

## POINT / Sun

$$\mathbf{l} = (\mathbf{p}_L - \mathbf{p}) / d$$

$$A(d) = \frac{1}{k_c + k_l d + k_q d^2}, \quad D = \max(\mathbf{n} \cdot \mathbf{l}, 0)$$

## SPOT / camera

$$C = \text{clamp}\left(\frac{\theta - \cos 18^\circ}{\cos 12^\circ - \cos 18^\circ}, 0, 1\right)$$

## DIRECTIONAL / fill

$$\mathbf{l} = -\text{normalize}(\mathbf{d}_L)$$



Frame 1

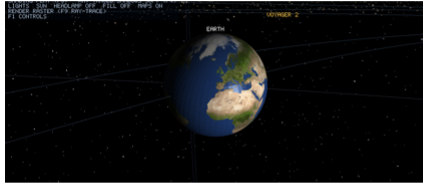
Frame 2

Frame 3

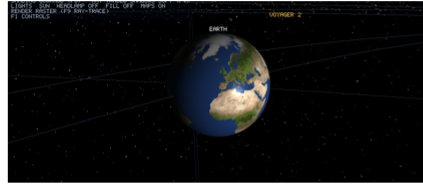
Camera motion carries the spotlight position and direction; the cone is  $12^\circ$ – $18^\circ$ .

# Five shading techniques, matched views

09 SHADING · Change the normal/evaluation stage while keeping geometry, texture and lights fixed



Flat



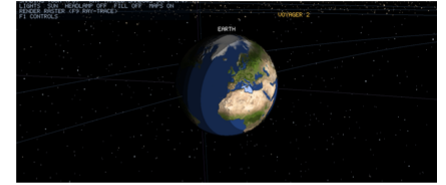
Gouraud



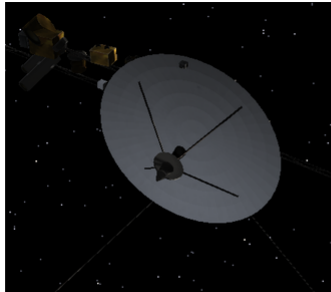
Phong



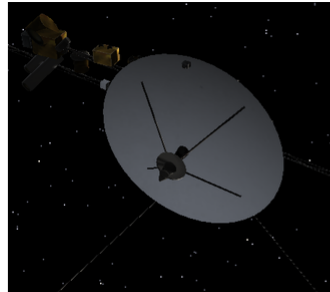
Blinn-Phong



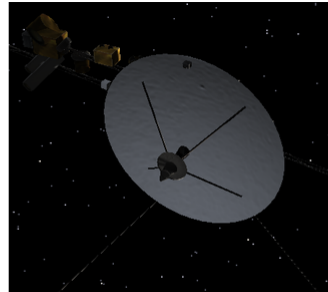
Toon



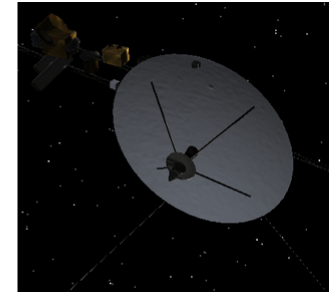
Flat



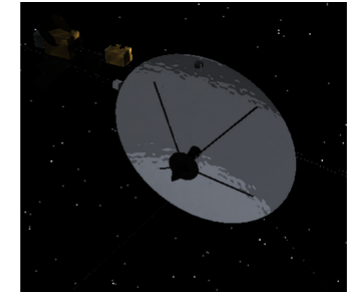
Gouraud



Phong



Blinn-Phong



Toon

## FLAT

Face normal

## GOURAUD

Vertex light

## PHONG

Pixel reflect

## BLINN

Pixel half-vector

## TOON

Bands + rim

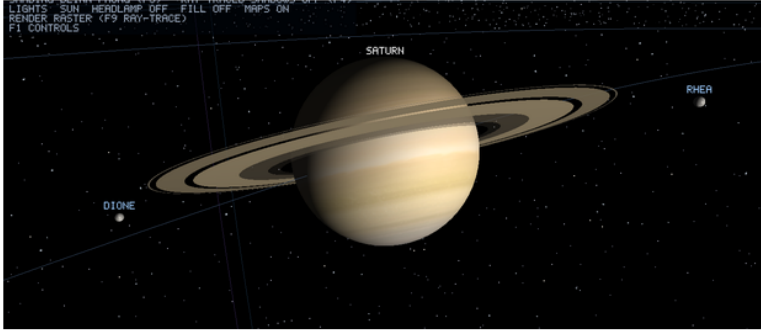
$$S_P = k_s \max(\mathbf{v} \cdot \mathbf{r}, 0)^p$$

$$S_B = k_s \max(\mathbf{n} \cdot \mathbf{h}, 0)^{2p}$$

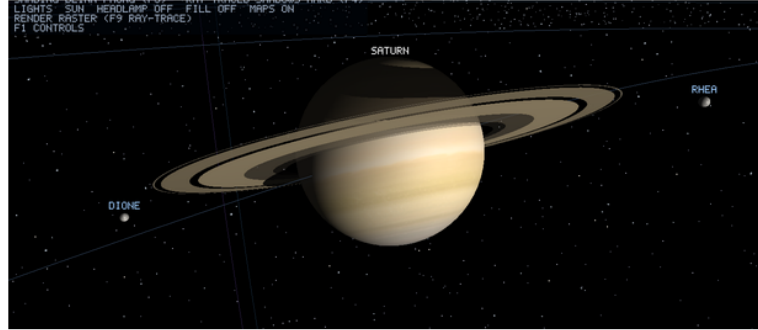
Toon: 1 / 0.62 / 0.30 / 0.06 bands; rim at  $\mathbf{n} \cdot \mathbf{v} < 0.22$ . Glow/unlit are separate materials.

# Visibility: eclipses, ring shadows and self-shadow

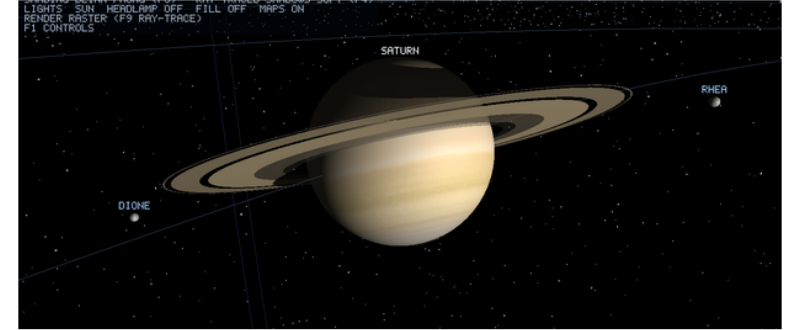
10 SHADOWS · Analytic Sun rays for bodies/rings; a depth map for the spacecraft



Off



Hard



Soft

$$V_{soft} = 1 - \frac{A_{overlap}}{\pi \alpha_{Sun}^2}, \quad V_{ring} = 1 - \text{opacity}$$

## BODIES + RINGS

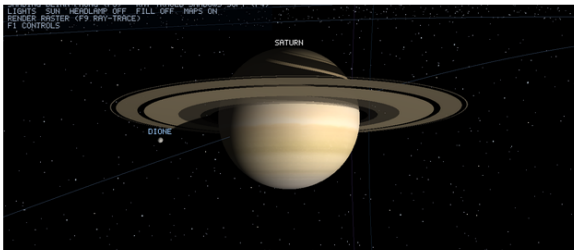
Analytic shadow rays: sphere / annulus.  
Finite Sun discs create a stable penumbra.

## VOYAGER SELF-SHADOW

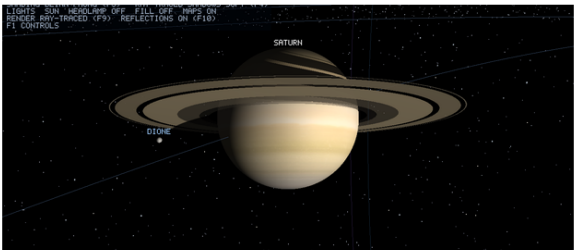
2048<sup>2</sup> depth map · 3×3 PCF comparisons.  
Bias avoids the surface shadowing itself.

# Ray tracing: nearest hit, light, reflection

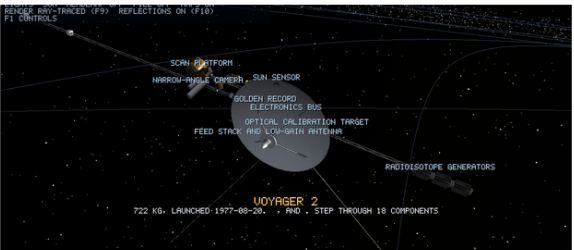
11 RAY TRACING · Analytic spheres/rings and a triangle BVH for the authored spacecraft



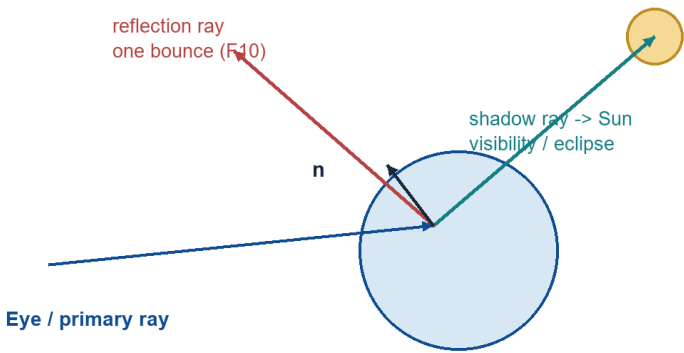
Saturn / raster



Saturn / traced



Voyager / traced BVH

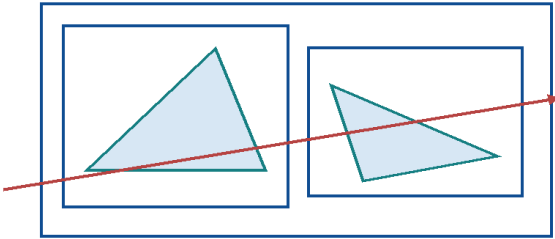


### NEAREST HIT

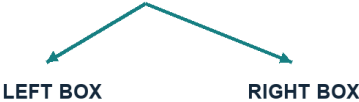
Sphere: quadratic roots  
Ring: plane + radius interval  
Voyager: triangles in BVH

Then evaluate texture, normal and Blinn-Phong. Translucent rings continue through up to four layers.

### SLAB TEST REJECTS EMPTY BOXES



### ROOT AABB



Near child first; closest t prunes far boxes.  
Leaves: up to 4 triangles.  
12-bin surface-area heuristic split.

5,828 triangles / 6,723 nodes / depth 24 / 32-entry traversal stack

CPU builds once. GPU texture buffers store nodes, vertices, normals, UVs and material indices.

$$\mathbf{p}(t) = \mathbf{o} + t\mathbf{d}, \quad t = -b \pm \sqrt{b^2 - f}$$

$$\mathbf{d}_r = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{n})\mathbf{n}$$

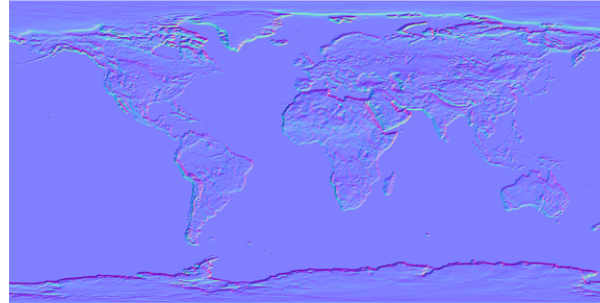
F9 hybrid Whitted view · 5,828 BVH triangles · up to 4 ring layers · F10 one mirror bounce

# Textures: colour, atlas, relief and specular mask

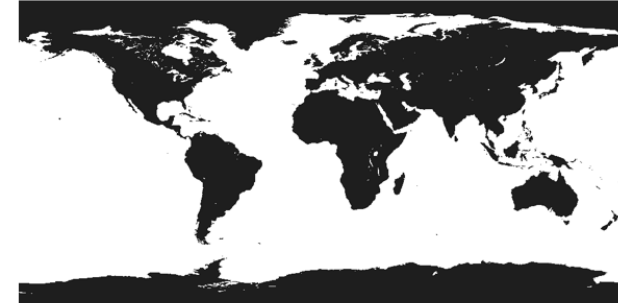
12 TEXTURES · The mesh supplies shape; image channels supply surface detail



Albedo / RGB colour



Sobel-derived normal / approximation



Blue-dominant ocean specular mask



Moon / maps off



Moon / maps on



**NASA atlas**  
**11 textured finishes**  
**+ copper material**

$$\mathbf{n}_t = \text{normalize}(-sS_x(H), -sS_y(H), 1)$$

$$uv' = uv \cdot zw + xy$$

F8 maps off/on · Sobel relief is inferred from colour · ocean mask controls highlights

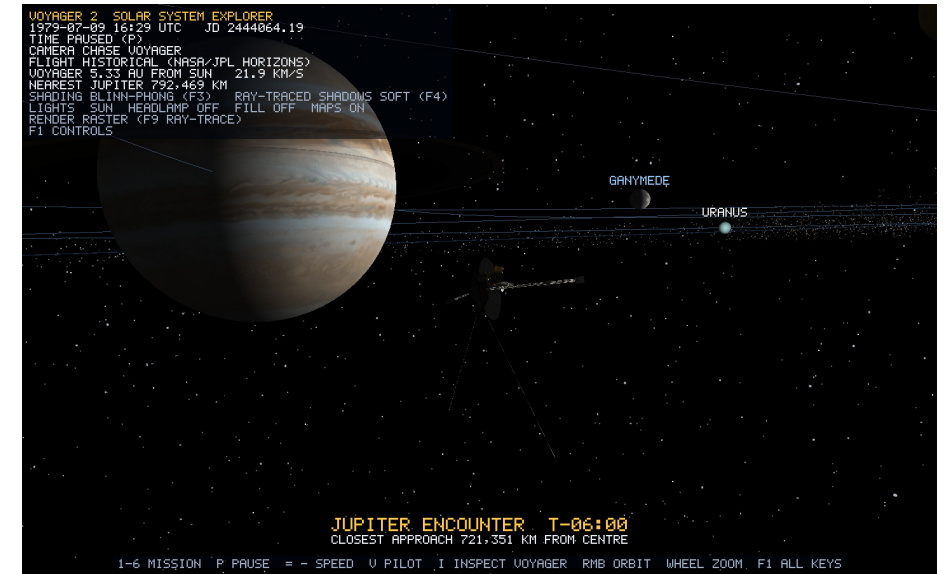
# Thank you

13 DISCUSSION · Questions about the project, its mathematics and its implementation

## From indexed triangles to an explorable mission.

Geometry · transforms · motion  
lighting · shading · ray tracing · textures

Verified: builds, scene checks and runtime captures.  
Measured raster views: 0.64–1.15 ms / RX 590.  
Explainable limits: compressed scale, visual moons,  
hybrid tracing and one reflection bounce.



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